

Assignment: Diabetes Prediction using Machine Learning

In this assignment, you will build a **Machine Learning classification model** using the provided **Diabetes Dataset** to predict whether a patient is likely to have diabetes based on various medical attributes.

Your goal is to implement an end-to-end **Machine Learning classification pipeline**, including data exploration, preprocessing, feature engineering, model building, training, evaluation, and performance analysis using **Python and Scikit-learn**.

This assignment will help you understand:

- End-to-end **Machine Learning workflow**
- Exploratory Data Analysis (EDA)
- Data preprocessing and handling missing values
- Feature selection and feature engineering
- Splitting data into training and testing sets
- Building and training **classification models**
- Understanding classification algorithms such as **Logistic Regression, Decision Trees, and Random Forest**
- Model evaluation using metrics such as **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix**
- Comparing model performance and selecting the most suitable model
- Understanding **overfitting and underfitting**
- Interpreting Machine Learning model results and drawing business/analytical insights

Dataset Description

The dataset contains patient health-related attributes that can be used to predict the likelihood of diabetes.

Some important columns include:

Column	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	Serum insulin level

BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree score
Age	Age of the patient
Outcome	Target Variable (0 = No Diabetes, 1 = Diabetes)

Assignment Tasks

You must complete the following stages as part of this Machine Learning assignment.

Part 1 — Data Preparation

Prepare and preprocess the dataset for Machine Learning model training.

Tasks

1. Load the dataset using **Pandas**.
2. Explore the dataset:
 - a. Shape of the dataset
 - b. Column names
 - c. Data types
 - d. Missing values, if any
 - e. Basic statistical summary
3. Separate the dataset into:
 - a. **Features (X)**
 - b. **Target Variable (y)**
4. Perform a **Train-Test Split**.
5. Perform appropriate preprocessing, if required:
 - a. Handle missing values
 - b. Encode categorical variables, if any
 - c. Scale/standardize numerical features where appropriate

Deliverables

- Dataset exploration summary
- Description of features and target variable
- Preprocessing steps performed
- Training and testing dataset sizes

Part 2 — Machine Learning Model Building

Build a suitable **classification model** using Scikit-learn.

Tasks

1. Import the required **Python and Scikit-learn libraries**.
2. Select and implement an appropriate classification algorithm, such as:
 - a. Logistic Regression
 - b. Decision Tree Classifier
 - c. Random Forest Classifier
3. Train the selected Machine Learning model using the training dataset.
4. Explain:
 - a. Why the selected algorithm is suitable for this problem
 - b. Key parameters used in the model
 - c. How the model makes predictions
5. If multiple algorithms are implemented, compare their performance and identify the best-performing model.

Deliverables

- Machine Learning model(s) implemented
- Explanation of the selected algorithm
- Key model parameters
- Comparison of models, if multiple algorithms are used

Part 3 — Model Training

Train the Machine Learning model using the prepared training dataset.

Tasks

1. Fit the selected model using the training dataset.
2. Generate predictions on the training dataset.
3. Evaluate the model's training performance.
4. Check for potential **overfitting or underfitting** by comparing training and testing performance.

Deliverables

- Model training output

- Training accuracy
- Training performance metrics
- Brief observation on overfitting/underfitting

Part 4 — Model Evaluation

Evaluate the performance of the trained Machine Learning model on the test dataset.

Tasks

1. Generate predictions on the test dataset.
2. Evaluate the model using:
 - a. **Accuracy Score**
 - b. **Precision**
 - c. **Recall**
 - d. **F1-Score**
 - e. **Confusion Matrix**
3. Analyze the confusion matrix and identify:
 - a. True Positives
 - b. True Negatives
 - c. False Positives
 - d. False Negatives
4. Compare the training and testing performance of the model.

Deliverables

- Test accuracy
- Precision, Recall, and F1-Score
- Confusion Matrix
- Comparison of training vs. testing performance
- Brief interpretation of the model results

Part 5 — Model Performance Analysis

Analyze the overall performance of the Machine Learning model and draw meaningful conclusions.

Tasks

1. Identify the most important performance metric for the diabetes prediction problem and explain why.
2. Discuss whether the model is performing well based on the evaluation results.

3. Identify potential limitations of the model.
4. Suggest at least **two ways to improve model performance**, such as:
 - a. Feature engineering
 - b. Hyperparameter tuning
 - c. Feature selection
 - d. Trying a different classification algorithm
 - e. Handling class imbalance

Deliverables

- Final model performance summary
- Key findings
- Model limitations
- Recommendations for improvement